

A probabilistic, network-based approach to voxel-based morphometry

G. Lynn Kurteff,^{1*} Grant M. Walker,² Siddarth Ramkrishnan,¹ Howard J. Rosen,¹ William W. Seeley,^{1,3} Maria Luisa Mandelli,¹ Zachary A. Miller,¹ & Maria Luisa Gorno-Tempini¹

Weill Institute for
Neurosciences | Fein Memory and
Aging Center

¹ Edward and Pearl Fein Memory and Aging Center, Department of Neurology, UCSF Weill Institute for Neurosciences, University of California, San Francisco; ² Department of Cognitive Sciences, University of California, Irvine; ³ Department of Pathology, University of California, San Francisco
* Corresponding author: lynn.kurteff@ucsf.edu

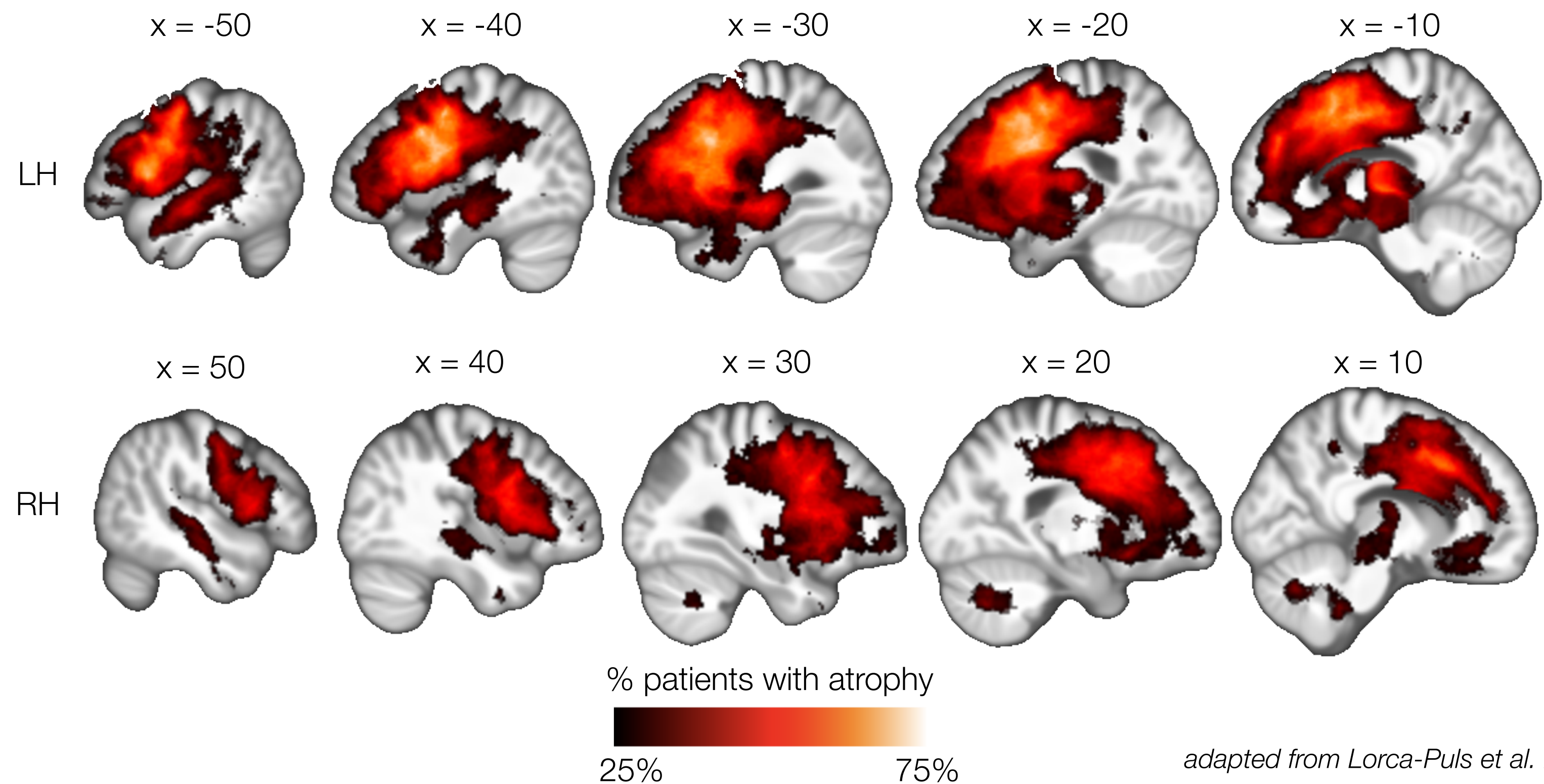
Background

- Voxel-based morphometry (VBM) is a popular metric for quantifying atrophic extent in neurodegenerative lesion-symptom mapping.¹
- Voxelwise null hypothesis statistical testing is the gold standard for LSM, but requires heavy correction due to number of comparisons.²
- Complex modeling can address shortcomings of traditional voxelwise LSM, but is prone to ad hoc parameter tuning (“researcher degrees of freedom”) and interpretability issues.³
- Classically, LSM is a single “winner-take-all” solution which may obscure diversity within a patient cohort and/or imaging dataset.
- Nonfluent variant primary progressive aphasia (nfvPPA)-spectrum disorders involve heterogeneous impairment to motor speech and grammar across participants.⁴
- nfvPPA-spectrum disorders are caused by several different FTLD pathologies which further complicates brain-behavior mapping.

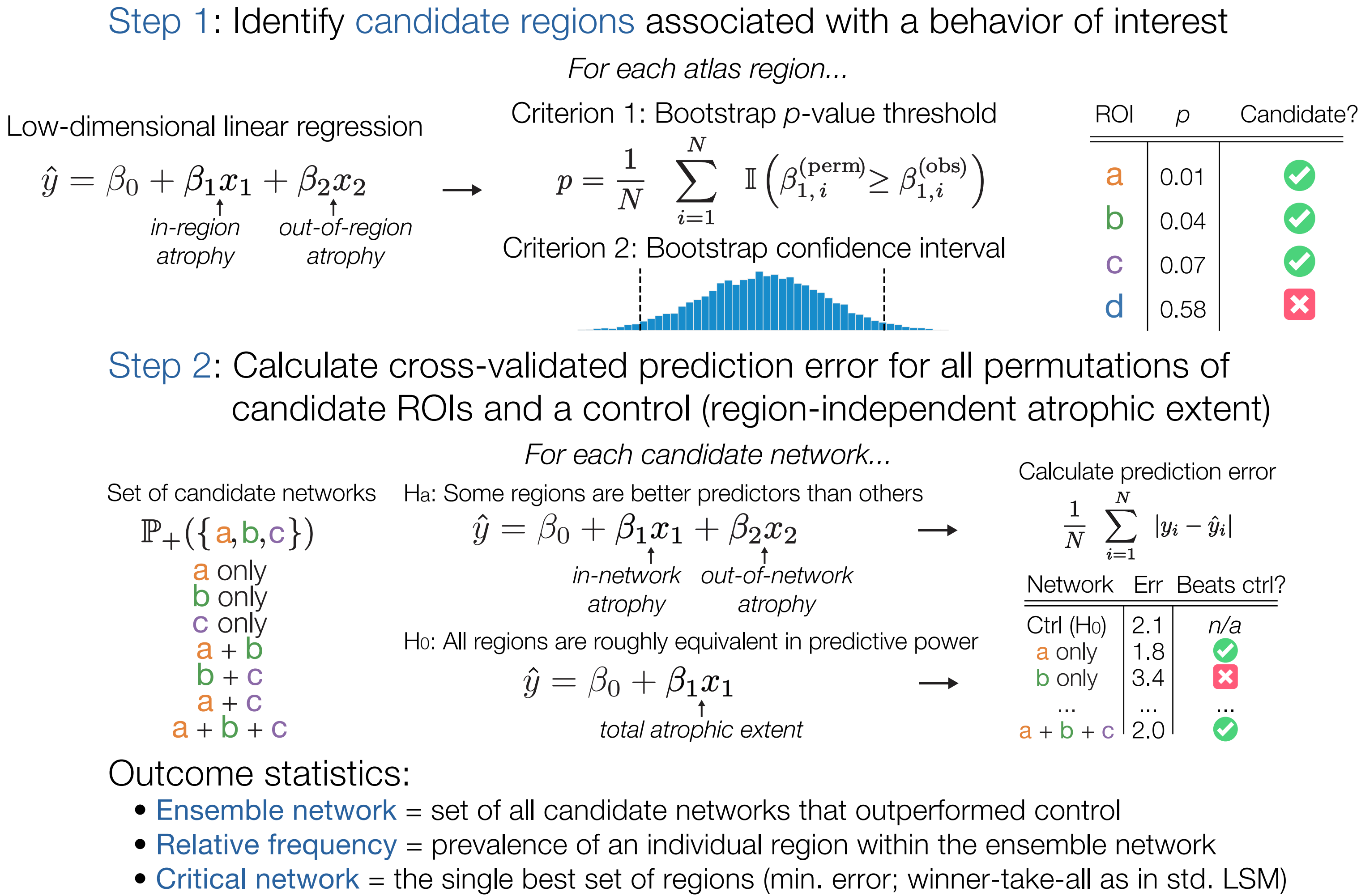
Goal: Validate a probabilistic alternative to traditional LSM that could be better-suited for the behavioral and neural heterogeneity of PPA.

The validation dataset

- Validating **critical network lesion-symptom mapping (CN-LSM)** against published work from our group (Lorca-Puls et al., 2024⁵).
- Participants: 104 cases of nfvPPA from the UCSF Fein Memory and Aging Center cohort with speech-language assessment
- Behavioral variables:
 - Apraxia of speech** and **dysarthria** severity rated on a 7-point scale by trained SLPs
 - Expressive agrammatism** index calculated using SALT transcript-derived morphosyntactic variables (MLU, % of utterances with errors, subordination index)
 - Receptive agrammatism** quantified from sentence comprehension task performance (sentence-to-picture matching)
- Atrophy quantified via VBM W-maps
 - W-maps are Z-scored volumetric changes to gray & white matter volume compared to healthy controls, adjusted for covariates
 - A cutoff of $W < -1.64$ used to binarize atrophy presence/absence



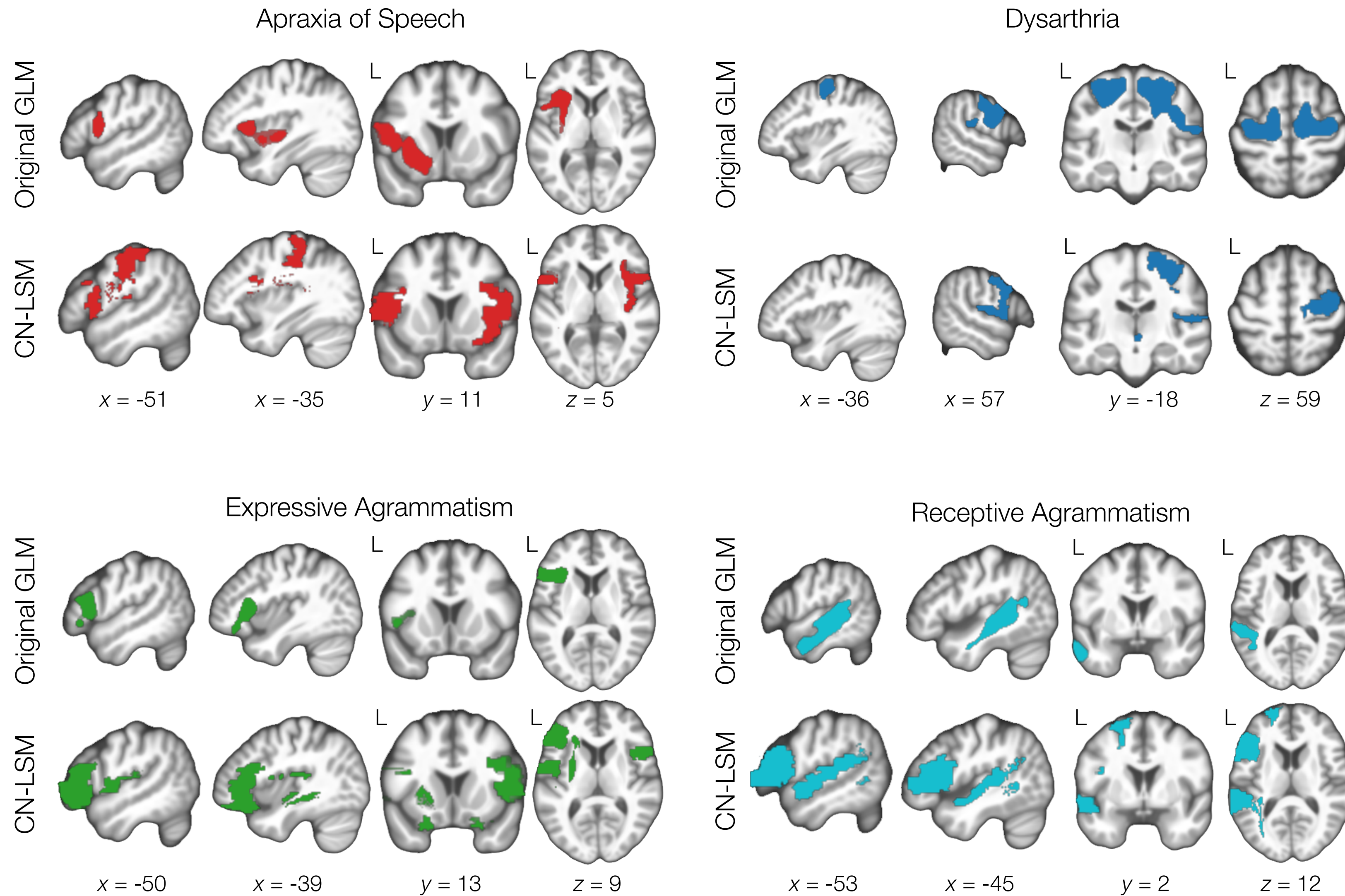
Critical Network Lesion-Symptom Mapping



For more info on this method, please see Grant Walker’s poster on CN-LSM in stroke! (Poster D54, Session D, Oct 1 4:30 to 6:30)

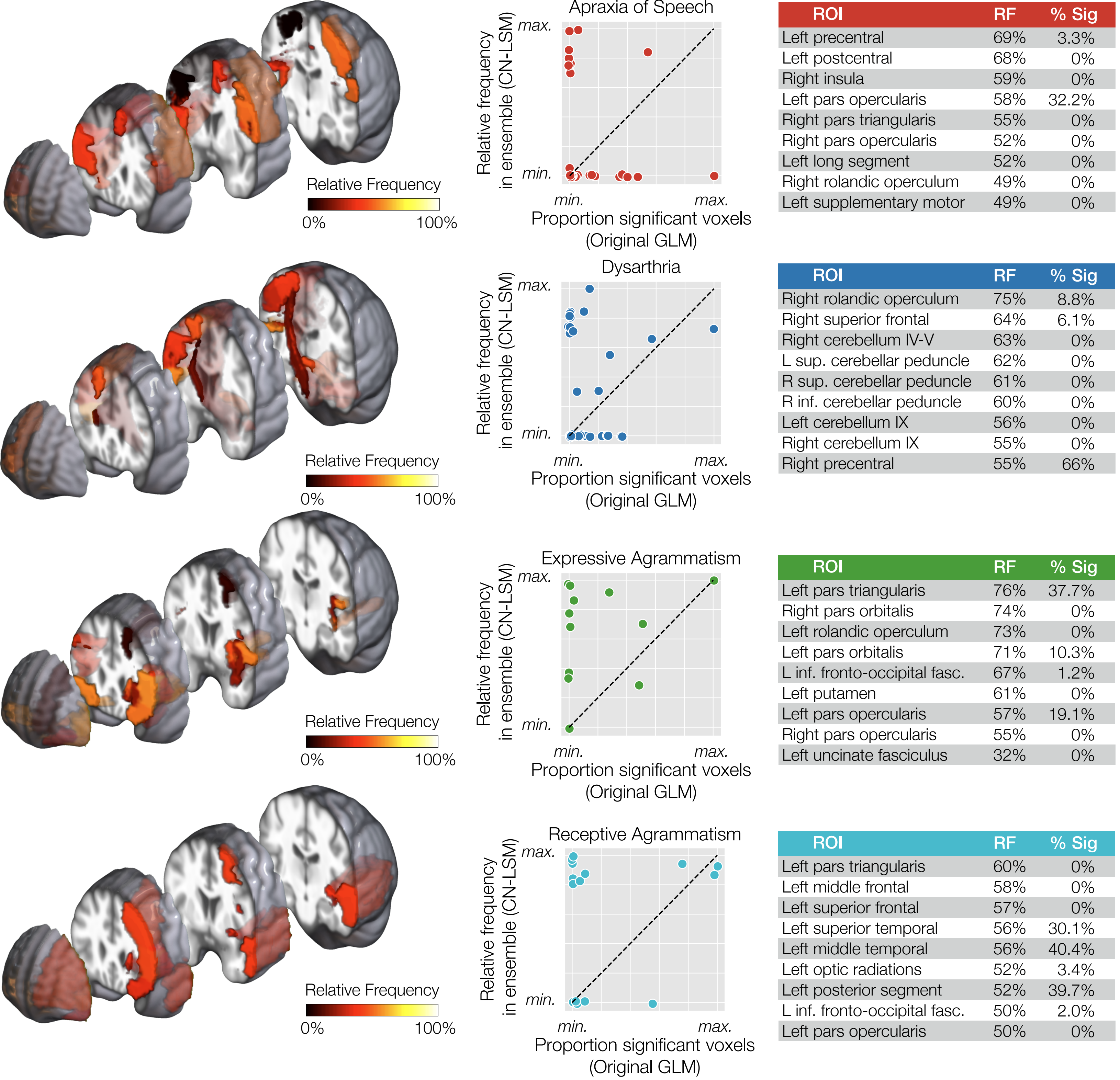
Comparing the critical networks

- Critical networks mapped domains of impairment in nfvPPA to similar regions as the original multivariate regression but with fewer statistical assumptions and comparisons.



Probabilistic insights

- Beyond a single optimal mapping, CN-LSM also allows investigation of the relative frequency of an individual region within the ensemble of networks
- CN-LSM ensemble frequency is not merely a proxy for in-ROI proportion of significant voxels in conventional LSM.



Discussion

- CN-LSM maps provide converging evidence to the original study while uncovering potential additional regions of interest.
- CN-LSM is suited for analysis of neurodegenerative imaging datasets.
- Future work aims to utilize CN-LSM in mapping domains of impairment in other PPA subtypes.

References and Acknowledgements

[1] Mechelli, A., Price, C.J., Friston, K.J., & Ashburner, J. (2005). Voxel-based morphometry of the human brain: methods and applications. *Curr Med Imaging Rev.*

[2] Sparber, C. (2020). Rethinking causality and data complexity in brain lesion-behaviour inference and its implications for lesion-behaviour modeling. *Cortex.*

[3] Walker, G.M., Kurteff, G.L., Hussain, A., Rorden, C., Bonilha, L., Fridriksson, J., & Hickok, G. (2026). Critical network lesion-symptom mapping. *Hum Brain Mapp.*

[4] Ilán-Gala, I., Lorca-Puls, D.L., Tee, B.L., Ezze, Z., de Leon, J., Miller, Z.A., Rubio-Guerra, S., Santos-Santos, M., Gómez-Andrés, D., Grinberg, L.T., Spina, S., Kramer, J.H., Wauters, L.D., Henry, M.L., Boxer, A.L., Rosen, H.J., Miller, B.L., Seeley, W.W., Mandelli, M.L., & Gorno-Tempini, M.L. (2024). Clinical dimensions along the non-fluent variant primary progressive aphasia spectrum. *Brain.*

[5] Lorca-Puls, D., Gajardo-Vidal, A., Mandelli, M.L., Ilán-Gala, I., Ezze, Z., Wauters, L.D., Battistella, G., Bogley, R., Rattasari, B., Licata, A.E., Battista, P., García, A.M., Tee, B.L., Lukic, S., Boxer, A.L., Rosen, H.J., Seeley, W.W., Grinberg, L.T., Spina, S., Miller, B.L., Miller, Z.A., Henry, M.L., Dronkers, N.F., & Gorno-Tempini, M.L. (2024). Neural basis of speech and grammar symptoms in non-fluent variant primary progressive aphasia spectrum. *Brain.*

[6] Yurganov, C., Smith, K.G., Fridriksson, J., & Rorden, C. (2015). Predicting aphasia type from brain damage measured with structural MRI. *Cortex.*

We thank our patients and their families for their participation in this research. This work was supported by the following National Institutes of Health grants awarded to M.L. Gorno-Tempini: R01NS05915, K24DC01554, R01AG075775, and P01AG019724.



Scan QR for poster PDF,
contact info, references